Order transfer through six for Williamson low-storage Runge–Kutta methods

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Abstract

Bazavov conjectured that replacing the additive updates in a Williamson low-storage Runge–Kutta method by exponential updates preserves its classical order for orders three, four and five. We prove this statement for every finite stage count and extend it through order six. The proof rests on an all-degree identity which expresses certain chronological products using only classical tableau data. If A, b are the associated classical coefficients, then for arbitrary vectors u, v and every $m \geq 1$,

$$t_m(u, v, \dots, v) = \frac{1}{m!} b^T R_v^{m-1} u, \qquad R_v = (b^T v) I - \operatorname{diag}(Av) + \operatorname{diag}(v) A.$$

A division-free row induction proves the identity, including degenerate coefficient lists. Its polarisation and the shuffle identities determine every required ordered tensor through total weight six. An integration homomorphism then shows that the classical order conditions force the exact ordered-flow coefficients. A fourth-order proof in conventional tableau notation gives an elementary entry point. Exact symbolic reduction checks all 197 ordered-tree coefficients through degree six. At weight seven the determining equations have three residual directions. Three explicit additional scalar tests suffice for seventh-order transfer, but their automatic satisfaction remains unproved. No unrestricted-order claim is made.

1. Problem and contribution

Many differential equations evolve within a constrained set of matrices: rotations remain orthogonal, for example, and quantum evolution remains unitary. An exponential update respects such a group constraint when its exponent lies in the corresponding Lie algebra. The difficulty is preserving the accuracy of an ordinary Runge–Kutta method after making this change. Matrix products do not commute, so merely reusing the same coefficients is generally insufficient. This paper identifies a structured family for which the additional order conditions follow automatically through order six.

Consider a sufficiently smooth matrix equation

$$Y'(t) = F(Y(t))Y(t).$$

Here $F(Y)$ lies in the Lie algebra of the matrix group under consideration. A natural geometric method updates Y by multiplication by exponentials. Its matrix products do not commute, so

classical Runge–Kutta order conditions generally require additional conditions to imply the same order on the group (Owren, 2006).

Williamson's two-register formulation is

$$K_i = U_i K_{i-1} + hf(y_{i-1}), \qquad y_i = y_{i-1} + V_i K_i, \quad 1 \le i \le s,$$

with $K_0 = 0$. Bazavov's corresponding lift is

$$Z_i = U_i Z_{i-1} + hF(Y_{i-1}), \qquad Y_i = \exp(V_i Z_i) Y_{i-1}, \quad Z_0 = 0. \tag{1}$$

The left-multiplication convention in (1) is fixed throughout. The last exponential appears on the left of the product, while the chronological order used in the pullback tensors runs from early rows to late rows.

Bazavov (2022, Section 4.3) conjectured order transfer for the higher-stage third-order and the fourth- and fifth-order methods. Bazavov (2026) still described the general claim as conjectural. Lystad (2026) proved the all-stage third-order case. We establish the explicitly conjectured orders three, four and five for every finite stage count, and extend the implication to order six. One tensor identity handles every coefficient list, while a small finite uniqueness argument accounts for the order boundary. The all-stage third-order theorem is prior work and is not claimed as new.

The practical consequence is precise: a developer who already has a Williamson tableau of classical order $p \le 6$ obtains a commutator-free Lie-group method of at least that order by applying (1). The theorem does not compare error constants, stable step sizes, or computational performance between particular methods.

More explicitly, two successive updates give

$$Y_2 = \exp(V_2 Z_2) \exp(V_1 Z_1) Y_0.$$

The latest exponential is on the left. The two-register description refers to the recurrence, not to all workspace needed to evaluate F or apply an exponential. The theorem concerns exact exponential updates; errors from an approximate exponential routine require their own control.

The proof has four steps. The low-storage recurrence forces an alternating row identity. That identity determines tensors whose arguments after the first are repeated. Polarisation and shuffle equations recover the remaining orderings through weight six. Finally, a continuous integration model identifies these expressions with the coefficients of the exact flow. Section 5 first shows the mechanism at order four without tree notation.

For use in a solver, the input is a verified classical Williamson coefficient list of order $p \le 6$, and the output is the recurrence (1) with consistency order at least p. This removes the need to solve additional noncommutative order constraints for that lift. Constructing the classical list, selecting its error constants and assessing its stability remain separate tasks. Yan (2017) supplies a thirteen-stage example of nonlinear order five, although its order for autonomous linear systems is eight; Bazavov (2025, Section 5.4) identifies its Williamson form. Printed decimal coefficients are not an exact algebraic certificate. We supply a rational fourth-order example below, but no sixth-order coefficient construction.

2. Coefficients and ordered tensors

Vectors e_i^T are the coordinate rows of $\mathbb{R}^s$. Define rows

$$w_0 = 0, \quad w_i = U_i w_{i-1} + e_i^T, \qquad \gamma_i = V_i w_i,$$

$$a_i = \sum_{j<i} \gamma_j, \qquad b^T = \sum_{i=1}^{s} \gamma_i. \tag{2}$$

The classical Butcher matrix A has row a_i; it is strictly lower triangular. Set

$$c = A\mathbf{1}, \quad C = b^T\mathbf{1}, \quad D = \operatorname{diag}(c), \quad R = CI - D + A. \tag{3}$$

No order or consistency assumption is imposed in this section. In particular, C need not equal one.

For vectors $u_1, \dots, u_m$, define the chronological tensor

$$t_m(u_1, \dots, u_m) = \sum_{1 \le i_1 \le \cdots \le i_m \le s} \frac{\prod_{r=1}^{m} (\gamma_{i_r} u_r)}{\prod_{j=1}^{s} n_j!}, \tag{4}$$

where n_j counts the occurrences of j among the indices. Equation (4) is Owren's tensor (24), with his chronological exponential rows β_j equal to γ_j. In particular,

$$t_2(u, v) = \sum_{i<j} (\gamma_i u)(\gamma_j v) + \frac{1}{2} \sum_i (\gamma_i u)(\gamma_i v). \tag{5}$$

Let $T_{2,r}$ denote the same tensor for the prefix $\gamma_1, \dots, \gamma_{r-1}$. This prefix is precisely the exponential representation of stage r.

For two rows the definition becomes

$$t_2(u, v) = \tfrac{1}{2}(\gamma_1 u)(\gamma_1 v) + (\gamma_1 u)(\gamma_2 v) + \tfrac{1}{2}(\gamma_2 u)(\gamma_2 v).$$

The halves come from choosing two factors from one exponential series. The mixed term chooses one factor from each row in early-to-late order. Reversing chronology replaces that mixed term by $(\gamma_2 u)(\gamma_1 v)$, which need not agree. This chronological convention is the pullback convention corresponding to the left-multiplication product displayed above.

3. A row identity

Lemma 1. For every pair of vectors u, v, every row in (2) satisfies

$$w_i((Av) \odot u - v \odot (Au)) = (a_i v)(w_i u) - (w_i v)(a_i u), \tag{6}$$

where $\odot$ denotes componentwise multiplication.

Proof. The claim for $i = 1$ is immediate. Suppose it holds for $i - 1$. The prefix changes by $a_i = a_{i-1} + V_{i-1} w_{i-1}$. Replacing a_{i-1} by a_i in the alternating expression on the right does not change its value, because the two added products cancel. The coordinate row e_i^T satisfies (6), since $(Au)_i = a_i u$ and $(Av)_i = a_i v$. Multiply the previous-row identity by U_i, add the coordinate-row identity, and use $w_i = U_i w_{i-1} + e_i^T$. No division occurs. □

For fixed v, put

$$C_v = b^T v, \qquad Q_i = \gamma_i v, \qquad d_i = C_v - a_i v, \qquad d_{s+1} = 0,$$

$$R_v = C_v I - \operatorname{diag}(Av) + \operatorname{diag}(v)A. \tag{7}$$

Then $d_i - d_{i+1} = Q_i$. Multiplying (6) by V_i gives the row identity

$$\gamma_i R_v = d_i \gamma_i + Q_i \sum_{j<i} \gamma_j. \tag{8}$$

Equivalently, $\Gamma R_v = K\Gamma$, where Γ has rows γ_i and

$$K_{ij} = \begin{cases} Q_i & j < i, \\ d_i & j = i, \\ 0 & j > i. \end{cases} \tag{9}$$

Neither matrix needs to be invertible. Taking $v = \mathbf{1}$ recovers $R_v = R$ in (3).

4. An all-degree contraction identity

For $k \geq 0$, write

$$H_k(x, y) = \sum_{r=0}^{k} x^{k-r} y^r.$$

Thus $(x - y)H_k(x, y) = x^{k+1} - y^{k+1}$, including when $x = y$.

Lemma 2. For every integer $m \geq 1$,

$$(\mathbf{1}^T K^{m-1})_i = H_{m-1}(d_i, d_{i+1}). \tag{10}$$

Proof. The case $m = 1$ is immediate. Multiplication by K gives at index i

$$d_i H_{m-1}(d_i, d_{i+1}) + \sum_{j>i} Q_j H_{m-1}(d_j, d_{j+1}).$$

The sum telescopes to d_{i+1}^m, because $d_{s+1} = 0$. Adding this term to the first expression gives $H_m(d_i, d_{i+1})$. □

Theorem 3. For every pair of real vectors u, v, every finite real Williamson coefficient list, and every integer $m \geq 1$,

$$\boxed{t_m(u, v, \ldots, v) = \frac{1}{m!} b^T R_v^{m-1} u.} \tag{11}$$

Proof. In (4), group the terms by the first index i and its number of repetitions r. The remaining arguments are v, so summing the later rows gives

$$t_m(u, v, \ldots, v) = \sum_i (\gamma_i u) \sum_{r=1}^{m} \frac{Q_i^{r-1} d_{i+1}^{m-r}}{r!(m-r)!}. \tag{12}$$

The binomial theorem, interpreted as a polynomial identity when $Q_i = 0$, identifies the inner sum as $H_{m-1}(d_i, d_{i+1})/m!$. Equations (8) and (10), together with $b^T = \mathbf{1}^T \Gamma$, yield

$$b^T R_v^{m-1} = \mathbf{1}^T K^{m-1} \Gamma = \sum_i H_{m-1}(d_i, d_{i+1}) \gamma_i.$$

Substitute into (12). □

The bilinear case is particularly simple:

$$2t_2(u,v) = (b^T u)(b^T v) + b^T((Au)\odot v - (Av)\odot u). \tag{13}$$

For $m = 2, u = c, v = \mathbf{1}$, the formula recovers Lystad's third-order moment identity. Bazavov's c-reflection work contains the related operator $I - D + A$. These are antecedents of the present argument; we do not claim novelty for that operator or for the earlier third-order result.

5. Fourth-order transfer

The eight classical conditions through order four are

$$\begin{array}{llll} b^T\mathbf{1} = 1, & b^T c = \frac{1}{2}, & b^T c^2 = \frac{1}{3}, & b^T Ac = \frac{1}{6}, \\ b^T c^3 = \frac{1}{4}, & b^T \operatorname{diag}(c) Ac = \frac{1}{8}, & b^T Ac^2 = \frac{1}{12}, & b^T A^2 c = \frac{1}{24}. \end{array} \tag{C4}$$

Powers of vectors in (C4) are componentwise. Owren (2006, equations 26a, 27a, 28a, 29a and 31a) proves that, after imposing (C4), the remaining conditions through order four are

$$t_2(c,\mathbf{1}) = \tfrac{1}{6}, \quad t_3(c,\mathbf{1},\mathbf{1}) = \tfrac{1}{24}, \quad t_2(c^2,\mathbf{1}) = \tfrac{1}{12}, \quad t_2(Ac,\mathbf{1}) = \tfrac{1}{24}, \quad \sum_r b_r T_{2,r}(c,\mathbf{1}) = \tfrac{1}{24}. \tag{L4}$$

The first is the order-three condition. Its other orientation follows from $t_2(c,\mathbf{1}) + t_2(\mathbf{1},c) = (b^T c)(b^T\mathbf{1})$.

Theorem 4. Every finite-stage Williamson method satisfying (C4) has a lift (1) of consistency order at least four.

Proof. Since $C = 1$, apply (11) with $m = 2$ and $u = c, c^2, Ac$. The resulting values are respectively

$$\frac{1}{2}(\tfrac{1}{2} - \tfrac{1}{3} + \tfrac{1}{6}) = \tfrac{1}{6}, \quad \frac{1}{2}(\tfrac{1}{3} - \tfrac{1}{4} + \tfrac{1}{12}) = \tfrac{1}{12}, \quad \frac{1}{2}(\tfrac{1}{6} - \tfrac{1}{8} + \tfrac{1}{24}) = \tfrac{1}{24}.$$

For the third-order tensor, expand $(I - D + A)^2$, keeping the matrix order:

$$6t_3(c,\mathbf{1},\mathbf{1}) = b^T(c - 2c^2 + 2Ac + c^3 - DAc - Ac^2 + A^2 c).$$

Equation (C4) gives

$$6t_3(c,\mathbf{1},\mathbf{1}) = \tfrac{1}{2} - \tfrac{2}{3} + \tfrac{1}{3} + \tfrac{1}{4} - \tfrac{1}{8} - \tfrac{1}{12} + \tfrac{1}{24} = \tfrac{1}{4}.$$

Finally apply the $m = 2$ formula separately at every stage prefix. Its total mass is c_r, so

$$2T_{2,r}(c,\mathbf{1}) = c_r(Ac)_r - (Ac^2)_r + (A^2 c)_r.$$

Weighting with b_r gives

$$\sum_r b_r T_{2,r}(c,\mathbf{1}) = \frac{1}{2}(\tfrac{1}{8} - \tfrac{1}{12} + \tfrac{1}{24}) = \tfrac{1}{24}.$$

Thus every condition in (L4) holds. Owren's complete order theory implies consistency order four. □

Under the standard local regularity assumptions for one-step convergence, the local defect is $O(h^5)$ and the global error on a fixed finite integration interval is $O(h^4)$. For nonautonomous

equations, adjoining time gives the usual autonomous formulation; the time component of the lifted constant frame reproduces the classical stage nodes.

6. Transfer through order six

The fourth-order argument illustrates the mechanism without trees. To handle orders five and six systematically, we combine Theorem 3 with the shuffle relations and the elementary-weight form of the classical order conditions.

6.1 Classical and ordered trees

Let τ be an ordinary, unordered rooted tree. Its number of vertices is $|\tau|$, and its tree factorial is

$$\gamma(\bullet) = 1, \qquad \gamma([\tau_1, \dots, \tau_k]) = |[\tau_1, \dots, \tau_k]| \prod_{j=1}^{k} \gamma(\tau_j).$$

Its stage elementary vector is

$$\Phi_\bullet = \mathbf{1}, \qquad \Phi_{[\tau_1,\dots,\tau_k]} = \bigodot_{j=1}^{k} A\Phi_{\tau_j},$$

and its output moment is $M_\tau = b^T \Phi_\tau$. The classical order-p conditions are

$$M_\tau = \frac{1}{\gamma(\tau)} \qquad (|\tau| \le p). \tag{C}$$

Every moment has weight $|\tau|$; weights add when moments multiply.

An ordered tree $t = (t_1, \dots, t_k)$ retains the order of its children. Write $g(t)$ for its numerical operator-series coefficient; its degree is $|t| - 1$, so the one-vertex ordered tree represents the identity operator. Owren's expansion gives

$$g(t) = t_k(g_{\mathrm{st}}(t_1), \dots, g_{\mathrm{st}}(t_k)), \tag{O}$$

where $g_{\mathrm{st}}(t_j)$ is the vector of coefficients at the evaluation-stage prefixes. The exact coefficients satisfy

$$e(\bullet) = 1, \qquad e((t_1, \dots, t_k)) = \frac{\prod_{j=1}^{k} e(t_j)}{\prod_{j=1}^{k} \left(\sum_{\ell=1}^{j} |t_\ell|\right)}. \tag{E}$$

Equation (E) is the iterated-integral form of Owren's exact-flow recursion (9). Our output $g(t)$ is his $g_{s+1,J}(t)$, and $e(t)$ is his $\alpha(t)/(|t|-1)!$. Owren's Theorem 2.5 and Corollary 2.6 imply that matching $g(t) = e(t)$ for $|t| \le p+1$ suffices for consistency order p throughout the stated problem class. Necessity is a universal order statement, not a claim about each individual equation, where special cancellations can occur.

The two tree conventions serve different purposes:

Object	Bookkeeping degree	Meaning
Ordinary tree τ	$\lvert\tau\rvert$	Weight of the classical moment M_τ
Ordered operator tree t	$\lvert t\rvert - 1$	Power of the step size in its output coefficient
Prefix conversion of an output polynomial of weight d	$d + 1$	Weight of its stage elementary vectors

The singleton ordered tree represents the identity operator, so it has degree zero. This accounts for the one-vertex shift; it is not an additional order condition.

6.2 Which tensors are determined?

The tensors (4) satisfy the shuffle identity

$$t_r(u_1, \dots, u_r)t_q(v_1, \dots, v_q) = \sum_{\sigma \in \mathrm{Sh}(r,q)} t_{r+q}(\sigma(u_1, \dots, u_r; v_1, \dots, v_q)). \tag{S}$$

Repeated words are counted with multiplicity. This follows either by partitioning the ordered integration simplex or by multiplying the defining products of exponential series.

Polarising Theorem 3 in its repeated vector supplies another relation. For a fixed first argument u, it determines the sum of all distinct permutations of any prescribed tail multiset. The right side is $1/m!$ times the sum of the corresponding operator products in the linear maps R_v, contracted with b^T. We denote this known equation by $P(u)$.

For example, the tail multiset $\{a, a, b\}$ gives

$$t_4(u, a, a, b) + t_4(u, a, b, a) + t_4(u, b, a, a) = \frac{1}{24} b^T (R_a^2 R_b + R_a R_b R_a + R_b R_a^2) u.$$

Extracting the coefficient of $x^2 y$ after substituting $v = xa + yb$ gives exactly the three distinct words displayed. Equivalently, mixed differentiation introduces a common factor 2!, which cancels on the two sides. In contrast, repeated interleavings in a shuffle equation retain their multiplicity.

Lemma 5 (finite tensor determination). For arguments chosen among the ordinary elementary vectors Φ_τ, with each argument assigned weight $|\tau|$, the polarised identities $P(u)$ and the shuffle identities determine every tensor of total weight at most six from classical moment polynomials.

Proof. Use induction on tensor length. A tensor of length one is a classical moment. A tensor with every argument equal follows from $t_m(v, \dots, v) = (b^T v)^m / m!$. For a prescribed multiset of arguments, regard the tensors for its distinct permutations as unknowns. In each shuffle equation, the factors on the right have smaller length and are already known. The right side of each $P(u)$ is a classical moment polynomial because R_v uses only A, b^T, and componentwise multiplication.

There is only one ordinary tree of weight one and only one of weight two. Consequently the nonconstant multiplicity patterns that can occur at total weight at most six are exactly those in the following table. Letters denote distinct arguments, independently of their order in the alphabet.

Pattern	Unknown permutations	A square set of known equations	Determinant
ab	2	P(a), P(b)	1
aab	3	P(a), P(b), S(a,ab)	1
abc	6	P(a), P(b), P(c), S(a,bc), S(ab,c), S(ac,b)	−1
aaab	4	P(a), P(b), S(a,aab), S(aa,ab)	1
aabb	6	P(a), P(b), S(a,abb), S(aab,b), S(a,bab), S(ab,ab)	2
aaaab	5	P(a), P(b), S(a,aaab), S(aa,aab), S(aaa,ab)	1

Here S(x,y) means the shuffle equation for the two indicated words. The integer coefficient matrices, with columns in lexicographic word order, are printed in Appendix A. Each is invertible over the rationals. This proves the induction. □

The assertion that the right side is a classical moment polynomial can be made explicit. If $\tau = [\tau_1, \dots, \tau_r]$ and $\sigma = [\sigma_1, \dots, \sigma_q]$, then

$$R_{\Phi_\sigma}\Phi_\tau = M_\sigma\Phi_\tau - \Phi_{[\tau_1,\dots,\tau_r,\sigma]} + \Phi_{[\sigma_1,\dots,\sigma_q,\tau]}. \tag{G}$$

This formula proves closure. Counting the weight of a scalar moment plus the weight of its stage tree, each term has weight $|\tau| + |\sigma|$. Applying R_{Φ_σ} therefore raises stage weight by $|\sigma|$. Repeated applications of (G), followed by contraction with b^T, expand every polarised right side into a rational polynomial in ordinary moments of the same total weight.

A polynomial in prefix moments is converted into a stage vector by replacing each monomial $M_{\tau_1}\cdots M_{\tau_k}$ with $\Phi_{[\tau_1,\dots,\tau_k]}$. Indeed, its i-th component is $\prod_j (a_i\Phi_{\tau_j})$, exactly the same monomial at stage i. The empty monomial becomes $\Phi_\bullet = \mathbf{1}$. Thus (O), Theorem 3 and Lemma 5 express each ordered coefficient of degree at most six as a rational polynomial in the classical moments. The polynomial for $g(t)$ is homogeneous of weight $|t| - 1$; converting it to a stage vector gives arguments of weight $|t|$.

For a concrete example, the ordered tree with two singleton children has output coefficient $t_2(\mathbf{1},\mathbf{1}) = \frac{1}{2}M_\bullet^2$. At evaluation stage i, this becomes $\frac{1}{2}c_i^2$, or $\frac{1}{2}\Phi_{[\bullet,\bullet]}$: output weight two becomes stage weight three. Truncating the recurrence after stage $i-1$ leaves every earlier component of an elementary vector unchanged because A is strictly lower triangular. Therefore the prefix substitution is valid at every stage, including degenerate coefficient lists.

6.3 Why the resulting coefficients are the exact ones

We now evaluate those moment polynomials under (C). This evaluation has a useful continuous interpretation. Replace the classical operations by

$$(Af)(x) = \int_0^x f(y)\,dy, \qquad b^T f = \int_0^1 f(x)\,dx,$$

and interpret componentwise multiplication as ordinary multiplication of functions. The elementary vector then becomes the polynomial

$$\Phi_\tau(x) = \frac{|\tau|}{\gamma(\tau)} x^{|\tau|-1},$$

so $b^T \Phi_\tau = 1/\gamma(\tau)$. This is precisely the substitution in (C), and it preserves every operation used in the reduction.

The continuous ordered tensors are iterated integrals. They satisfy (S) and the same repeated-tail formula. To verify the latter directly, write

$$\rho(x) = \int_x^1 v(y)\,dy, \qquad (R_v u)(x) = \rho(x)u(x) + v(x)\int_0^x u(y)\,dy.$$

Changing integration order gives, for every nonnegative integer j,

$$\int_0^1 \rho(x)^j (R_v u)(x)\,dx = \frac{j+2}{j+1}\int_0^1 \rho(x)^{j+1} u(x)\,dx.$$

Iterating yields

$$\frac{1}{m!}\int_0^1 R_v^{m-1} u(x)\,dx = \frac{1}{(m-1)!}\int_0^1 u(x)\rho(x)^{m-1}\,dx,$$

which is the iterated integral with first argument u and m−1 subsequent arguments v. No positivity or monotonicity assumption is needed.

Theorem 6 (order transfer through six). Let $1 \le p \le 6$. For every finite stage count and every real Williamson coefficient list satisfying the classical order conditions (C), the lift (1) has consistency order at least p. In particular, it resolves the order-3, order-4 and order-5 claim explicitly posed in Bazavov's Section 4.3 and extends the implication to order six.

Proof. Induct over ordered-tree size in (O). The stage coefficient for a child is the prefix version of the classical moment polynomial already constructed for that child. A prefix moment evaluates continuously to $x^{|\tau|}/\gamma(\tau)$. Homogeneity therefore makes the child stage coefficient $e(t_j)x^{|t_j|-1}$. Lemma 5 determines the root tensor uniquely using repeated-tail and shuffle equations. Continuous iterated integrals satisfy those same equations. Hence evaluation under (C) gives the corresponding continuous iterated integral, whose value is exactly (E). Every needed moment has weight at most $|t|-1 \le p$, so all substitutions are warranted by the assumed classical order conditions. Thus $g(t) = e(t)$ for every $|t| \le p+1$. Owren's completeness theorem gives consistency order p. □

For the analytic convergence consequence, suppose the exact solution and all sufficiently small-step numerical trajectories lie in a common neighbourhood on a fixed interval $[0, T]$, with bounded derivatives through the orders needed for the expansion and local Lipschitz bounds there. The smooth one-step map then has a Lipschitz bound $1+Lh$, and the local defect is uniformly $O(h^{p+1})$. Iterating $e_{n+1} \le (1+Lh)e_n + Kh^{p+1}$, with exact initial data and $nh \le T$, gives global error $O(h^p)$. These analytic assumptions are separate from the algebraic order conditions. The same estimate permits an inexact step map whose uniform difference from the exact lifted step is $O(h^{p+1})$; it does not cover arbitrary exponential approximations. In the nonautonomous case, adjoining an additive time coordinate with constant vector field reproduces the classical stage times on the extended state space.

6.4 A precise boundary at weight seven

The tensor multiset (a, a, a, b, b) has ten permutations. The complete collection of its polarised repeated-tail and proper shuffle equations has rank nine. With a the weight-one argument **1** and b the weight-two argument c, its total weight is seven. Appendix A gives a nonzero null vector and a nonzero minor. Our identities therefore leave one tensor direction undetermined at the next order. The script reports this rank deficiency and stops. It is a boundary of this proof method, not a counterexample to seventh-order transfer.

6.5 Three explicit tests at order seven

The missing direction just exhibited is not the only new family at weight seven. A full enumeration gives two further deficient families, involving the two ordinary vectors of weight three. This leads to a conditional extension which makes the next-order problem finite and explicit.

Theorem 7 (a sufficient order-seven criterion). Suppose a finite real Williamson coefficient list satisfies all classical order-seven conditions. If

$$t_5(\mathbf{1}, \mathbf{1}, c, \mathbf{1}, c) = \frac{1}{280}, \qquad t_4(\mathbf{1}, c, \mathbf{1}, c^2) = \frac{1}{84}, \qquad t_4(\mathbf{1}, c, \mathbf{1}, Ac) = \frac{1}{168}, \tag{C7+}$$

then its lift (1) has consistency order at least seven.

Proof. Every child of a degree-seven ordered coefficient has degree at most six. Its prefix coefficient is therefore a linear combination of ordinary elementary vectors by the preceding reduction. The root tensors have total weight seven. Every factor of a proper shuffle product has strictly smaller total weight and is already determined. The polarised repeated-tail right sides are classical moment polynomials of weight seven. After imposing the classical order-seven conditions, the difference between the actual tensor values and their continuous counterparts consequently lies in the nullspace of the corresponding determination matrix.

There are $1, 1, 2, 4, 9, 20, 48$ ordinary rooted trees of weights one through seven. Enumerating multisets of these arguments with total weight seven gives the following complete list; families distinguish the actual ordinary-tree arguments, whereas patterns retain only multiplicities.

Pattern	Number of families	Rank / number of words
a	48	1/1
ab	37	2/2
aab	13	3/3
abc	5	6/6
aaab	5	4/4
aaaab	2	5/5
aaaaab	1	6/6
aaaaaaa	1	1/1
aaabb	1	9/10
aabc	2	11/12

These counts can be reproduced by listing the partitions of seven and assigning ordinary trees to each part. The matrices use exactly the first-letter polarisation rows and all proper-shuffle rows defined in Lemma 5. The full-rank cases through length five are already covered there. For aaaaab, the rows $P(a), P(b), S(a, aaaab), S(aa, aaab), S(aaa, aab), S(aaaa, ab)$ give an invertible six-by-six system. Constant-word tensors are determined directly.

The aaabb family is $(\mathbf{1}, \mathbf{1}, \mathbf{1}, c, c)$. Its null vector is printed in Appendix A and has nonzero aabab coordinate. The first condition in (C7+) therefore removes its one free direction. The aabc pattern occurs twice, as $(\mathbf{1}, \mathbf{1}, c, c^2)$ and $(\mathbf{1}, \mathbf{1}, c, Ac)$. In lexicographic word order

$$(aabc, aacb, abac, abca, acab, acba, baac, baca, bcaa, caab, caba, cbaa)$$

its nullspace is spanned by

$$(0, 0, -1, 1, 1, -1, 1, -1, 0, -1, 1, 0)^T.$$

Exact elimination of the polarisation and shuffle matrix gives rank eleven. The abac coordinate is nonzero, so the remaining two conditions remove both free directions. The small matrices and this exhaustive family count are independently checkable by `code/order7_criterion.py`.

The constants follow directly from the continuous model. The iterated integral of $x_i^{w_i-1}$ over $0 < x_1 < \cdots < x_m < 1$ is

$$\left[\prod_{j=1}^{m}(w_1 + \cdots + w_j)\right]^{-1}.$$

Weights $(1, 1, 2, 1, 2)$ give $1/280$, and weights $(1, 2, 1, 3)$ give $1/84$. Since $Ac = x^2/2$ in that model, the third value is $1/168$. Every root tensor now agrees with its exact continuous value, and the ordered-tree sufficiency argument proves the result. □

At the level of matching the complete ordered series, the three tests are also necessary: the required child-stage coefficients are $\mathbf{1}, c, c^2/2, Ac$, so the three tensors occur as root coefficients, with a factor $1/2$ in the second case. This is a coefficient-level observation, not a necessity assertion for every special differential equation. We have not proved that (C7+) follows from the classical order-seven conditions, that its three conditions are independent on genuine Williamson lists, or that a seventh-order list exists. The unconditional transfer theorem remains bounded by six.

7. Exact checks and scope

The exact symbolic certificate records all 197 ordered-tree coefficients through degree six, including the degree-zero identity coefficient, as polynomials in ordinary moments. Its generator uses rational algebra, the polarised form of Theorem 3 and shuffle equations; it uses no numerical fitting. The independent checker imports only the Python standard library and does not import the generator. It reads the emitted formulas and verifies all 87 polarisation equations, 212 shuffle equations, 47 nontrivial family ranks, 197 root-recursion substitutions and 197 exact-flow values. It also independently enumerates the six allowed multiplicity patterns and checks the matrices in Appendix A. Thus the certificate can be checked without trusting the solver that produced it.

A separate verifier reconstructs the tableau directly from the low-storage recurrence. It performs 1,360 exact rational checks of the row and tensor identities on 20 coefficient lists, including degenerate lists and lists with negative coefficients, with tensor degree up to eight. It independently expands the stage exponentials on four arbitrary rational coefficient lists and compares all 197 emitted coefficient polynomials, giving 788 further exact comparisons. It also checks the eight classical and five ordered conditions for the rational five-stage fourth-order method in Bazavov (2025, Table 7):

$$U = (0, -1, -1, -11, \tfrac{1}{10}), \qquad V = (\tfrac{1}{2}, \tfrac{2}{3}, -\tfrac{1}{2}, -\tfrac{1}{10}, \tfrac{1}{6}). \tag{M4}$$

The direct ordered-tree expansion verifies every coefficient through degree four for (M4), with the first defects at degree five. A nonlinear noncommuting matrix check uses

$$F(Y) = X + Y_{12}W, \qquad X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad W = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}, \qquad Y(0) = I.$$

Direct truncated power-series arithmetic gives the first local-error term for (M4) at degree five. Reversing the multiplication convention gives a nonzero defect already at degree three, providing a negative control. These computations check transcription, orientation and concrete consequences; the all-stage theorem rests on the proofs in Sections 3–6.

The theorem imposes no positivity, monotone-node, distinct-node, or nonzero-coefficient requirement. It therefore covers both the regular coefficient parameterisations in the source literature and their direct-recurrence degeneracies. It does not assert that all classical Runge–Kutta methods admit Williamson storage. The theorem is an implication for every coefficient list satisfying the stated classical order conditions. It makes no existence claim for a prescribed order and stage count. The general repeated-tail tensor identity holds beyond order six, but those identities and shuffle equations alone cease to determine every required tensor at weight seven.

8. Prior-art boundary

The search, completed on 19 October 2026, inspected the originating conjecture, Owren's complete order conditions, Bazavov's 2025 c-reflection preprint and 2026 coefficient paper, and Lystad's August 2026 proof. Exact formulas, the operator $I - D + A$, low-storage aliases, Lie–Butcher terminology, and recent resolution searches were checked. The all-stage transfer theorem at orders four, five and six and the all-degree repeated-tail contraction identity were not found in that bounded corpus. This is evidence about the inspected corpus, not a proof of historical priority. The detailed queries and contribution-by-contribution decisions appear in `novelty_report.md`.

9. Conclusion and availability

The Williamson recurrence turns a restricted family of noncommutative tensor products into classical moment data at every degree. Through weight six, the remaining products are uniquely determined by polarisation and shuffles, proving automatic transfer of the stated classical orders. At weight seven the reduction leaves three scalar tests. Establishing whether those tests are automatic, or finding a genuine violating coefficient list, remains open.

The principal result is the written theorem, not a numerical trend. The exact certificate checks transcription and finite algebra; the rational and noncommuting examples check conventions and selected consequences. None establishes external peer review, formal proof-assistant verification, historical priority, or a performance advantage.

Data and code availability. The accompanying versioned source-and-evidence package contains this manuscript, the order-six certificate, the order-seven finite criterion checker, the independently implemented checks and `AI_INDEX.md`, which identifies the replay entry point and the scope of every claim. The source repository is https://github.com/ipitchford/williamson-order-transfer and the versioned archive is https://doi.org/10.5281/zenodo.23501588. The proof above is self-contained apart from the cited order-condition framework; no private review is required to interpret it.

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Appendix A. Finite integer certificates

For each pattern, columns list the distinct words in lexicographic order. Rows are the equations listed in Lemma 5, in that order. A row $P(a)$ has coefficient one in every word starting with a and zero elsewhere; a row $S(x, y)$ counts the interleavings of x and y, retaining multiplicities.

For ab, use columns (ab, ba):

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \det = 1.$$

For aab, use columns (aab, aba, baa):

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix}, \qquad \det = 1.$$

For abc, use columns $(abc, acb, bac, bca, cab, cba)$:

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{pmatrix}, \qquad \det = -1.$$

For aaab, use columns $(aaab, aaba, abaa, baaa)$:

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 3 & 1 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{pmatrix}, \qquad \det = 1.$$

For aabb, use columns $(aabb, abab, abba, baab, baba, bbaa)$:

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 2 & 1 & 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 4 & 2 & 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \det = 2.$$

For aaaab, use columns $(aaaab, aaaba, aabaa, abaaa, baaaa)$:

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 4 & 1 & 0 & 0 & 0 \\ 6 & 3 & 1 & 0 & 0 \\ 4 & 3 & 2 & 1 & 0 \end{pmatrix}, \qquad \det = 1.$$

For the weight-seven boundary aaabb, the columns are

$$(aaabb, aabab, aabba, abaab, ababa, abbaa, baaab, baaba, babaa, bbaaa).$$

Every polarised-tail and proper-shuffle row annihilates the nonzero column vector

$$(0, 1, -1, -3, 4, -1, 2, -3, 1, 0)^T.$$

Conversely, take the nine rows

$$P(a), P(b), S(a, aabb), S(aa, abb), S(aaab, b), S(a, abab), S(aa, bab), S(aab, ab), S(aaba, b).$$

Deleting the second column gives a nine-by-nine minor of determinant one. These facts certify rank exactly nine. They concern the determining equations, rather than the attainability of that remaining direction by Williamson coefficients.

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